Data Mining Project Proposal

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1. Project Title

Analyzing Socioeconomic and Lifestyle Determinants of Health Insurance Coverage and Diabetes:
A Data Mining Project Based on NHANES 2021–2023

2. Problem Statement

Health inequality remains a persistent challenge in the United States, where disparities in income, education, and access to care continue to influence both the availability and quality of healthcare services. Despite the expansion of health insurance coverage in recent years, variations in insurance type, such as public, private, dual, or none, reflect deeper socioeconomic divisions. These structural inequalities often shape individuals' ability to access preventive care, manage chronic diseases, and maintain overall well-being. At the same time, lifestyle behaviors such as sleep, exercise, and smoking are known to interact with social determinants, jointly contributing to disparities in chronic conditions such as obesity, hypertension, and diabetes. Understanding how socioeconomic, behavioral, and health factors interact to produce these inequalities is crucial for improving both healthcare access and health outcomes across diverse populations.

This project aims to investigate how socioeconomic characteristics, lifestyle factors, and physiological indicators collectively influence both health insurance coverage and chronic disease risk. Using data from the National Health and Nutrition Examination Survey (NHANES) 2021–2023, we employ data mining techniques to extract meaningful patterns, construct predictive models, and identify distinct health population profiles. By moving beyond descriptive analysis, this project seeks to uncover interpretable relationships and predictive insights that illuminate the mechanisms linking socioeconomic disparities to health outcomes. To achieve this aim, we define three main objectives:

First, we aim to identify underlying patterns between socioeconomic variables and insurance coverage types using association rule mining. Specifically, we examine how combinations of income, education with insurance status (public, private, both, or uninsured). In parallel, we explore how behavioral and physiological factors, such as BMI and sleep duration, relate to chronic disease risk, particularly diabetes. Algorithms including Apriori or FP-Growth will be utilized to generate interpretable association rules, evaluated through Support, Confidence, Lift, Kulc, and Imbalance Ratio metrics.

Second, we aim to validate these findings and develop predictive models using a set of classification algorithms. Two prediction tasks are defined: (1) predicting insurance type based on socioeconomic features; and (2) predicting diabetes risk using combined socioeconomic attributes and insurance status, physiological indicators. We will implement and compare Decision Tree, Random Forest, XGBoost classifiers. Model performance will be evaluated through standard metrics such as Accuracy, Precision, Recall, F1-score, Confusion Matrix, and ROC-AUC, testing the models' ability to capture interactive effects that drive health outcomes.

Third, we aim to identify subpopulations with distinct health and lifestyle patterns using clustering analysis. Through K-Means or Hierarchical Clustering, we seek to uncover meaningful population segments, such as "Healthy and Insured," "At-Risk Publicly Insured," or "Uninsured and Unhealthy." Cluster quality will be assessed using the Silhouette Score, facilitating interpretation of the overall population structure and highlighting groups that warrant targeted health interventions.

The potential applications of this research extend well beyond scientific understanding. The patterns and predictive insights derived from this project can inform public health initiatives and evidence-based policymaking, particularly in identifying uninsured or high-risk populations who could benefit from targeted health education, preventive care, or subsidized insurance programs. Moreover, the findings can assist healthcare providers and policymakers in designing more equitable intervention strategies and allocating resources more efficiently. By uncovering the data-driven mechanisms that underlie health disparities, this project contributes not only to the advancement of data science methodologies, but also to the broader societal goal of promoting fairness, inclusivity, and accessibility within healthcare systems.

3. Dataset Description

- **Source of Data:** The dataset used in this project is derived from the *National Health and Nutrition Examination Survey (NHANES August 2021-August 2023)*, a nationally representative survey conducted by the U.S. Centers for Disease Control and Prevention (CDC) and the National Center for Health Statistics (NCHS), publicly available through the official NHANES data portal (`https://wwwn.cdc.gov/nchs/nhanes/continuousnhanes/default.aspx`).

- **Type of Data:** The dataset is *structured tabular data* consisting of numeric and categorical variables collected through cross-sectional survey. Each row represents an individual participant identified by a unique sequence number (SEQN), and the dataset covers multiple NHANES modules:
  - Demographics Data (DEMO: age, gender, race, education, income-to-poverty ratio)
  - Examination Data (BMX, BPXO: physical measurements such as BMI, sbp, dbp)
  - Laboratory Data (AGP, GLU: clinical indicators (e.g., a1c, glucose))
  - Questionnaire Data (DIQ, HIQ, HUQ, PAQ, SLQ, SMQ, ect.: self-reported lifestyle behaviors, health conditions, insurance coverage (e.g., smoking, physical activity, sleep hours, diabetes, insurance type))
- **Key Features/Attributes:** The combined dataset provides a comprehensive view of socio-economic status, health indicators, lifestyle behaviors, and health insurance status. Representative attributes include:
  - *Demographics: age, gender, education, income_poverty_ratio*
  - *Health indicators: bmi, diabetes_self_report*
  - *Lifestyle:smoking_current, smoking_ever, physical_activity_freq, sleep_hours*
  - *Insurance status: insurance_type*
- **Data Size and Format:** The raw data originates from 17 individual NHANES survey component files in XPT (SAS transport) format. These files were merged using unique respondent identifiers and decoded based on official NHANES codebooks. The combined dataset initially consisted of 11,933 observations and 154 variables. After feature selection and aggregation, the analytical dataset was refined to approximately 11,933 records and 21 key features and exported as CSV(`nhanes.csv`) files for analysis in Python or R environments. Missing values were analyzed and retained for appropriate handling during model development. This format supports a wide range of analytical tasks, including association rule mining, classification modeling, and clustering analysis.

4. Algorithm choices/ideas

This project adopts a stepwise data mining framework: association rule discovery → classification → population risk segmentation (clustering) —to analyze how socioeconomic factors relate to health insurance type, how lifestyle behaviors, physiological indicators, and insurance choices jointly contribute to diabetes, and to further identify vulnerable high-risk subpopulations.

Association Rule Mining: Algorithms such as *Apriori* or *FP-Growth* will be applied to uncover statistically meaningful frequent patterns across the dataset, focusing on two key dimensions:

- Socio-economic characteristics → Insurance type
  e.g., income, education associated with public vs. private coverage patterns

- Lifestyle + physiological indicators → diabetes risk
  examining interactions among physical activity, sleep duration, BMI, in relation to disease risk

Classification: Supervised learning models including *Decision Tree*, *Random Forest* and *XGBoost* will be trained to provide precise individual-level classification predictions, further validating the broad patterns identified in the rule-mining stage:

- Socio-economic factors → Insurance type prediction
  assessing access differences across demographic and economic groups

- Insurance + physiological + socio-economic factors → Diabetes risk prediction
  testing how insurance status, physiological indicators, and socio-economic factors jointly predict the risk of diabetes

Clustering: Unsupervised algorithms such as *K-Means* or *Hierarchical Clustering* will be used to employed to segment individuals into distinct lifestyle-health profiles, identifying potential diabetes high-risk clusters with a focus on uninsured individuals and generating insights to inform public-health and policy strategies.

5. Model evaluation

Model performance will be evaluated across predictive accuracy, clustering quality, and interpretability.

Evaluation Metrics:

- Association Rules: *Support, Confidence, Lift, Kulc, Imbalance Ratio*;

- Classification: *Accuracy, Precision, Recall, F1-score, ROC-AUC, Confusion Matrix*;

- Clustering: *silhouette score*;

Validation Methods: Data will be divided into the training (80%) and testing (20%) subsets. A *k-fold cross-validation* (typically $k = 5$) will be used to ensure model robustness and mitigate overfitting.

6. References

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